

Grounds of Restriction and the Governance of Disclosure

in Generative Relational Academic Infrastructure

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Abstract

An infrastructure that records how understanding changes must be able to withhold. This paper asks on what grounds, and establishes that the answer determines not only whether a record is restricted but what may be withheld and what must still be disclosed. Four grounds are distinguished and shown to be irreducible to one another: rivalry, where a resource admits limited simultaneous use; hazard, where propagation of the content creates risk; exploratory vulnerability, where premature exposure suppresses the work; and appropriability, where excludability is required to fund production. Each restricts a different object and leaves a different residue, and the paper argues that a residue rule is what prevents a ground from justifying total silence. It then treats three difficulties the grounds create. Moving exploration into simulated environments relocates the governance problem from the allocation of instruments to the specification of models and benchmarks, and removes nothing. The component of a trajectory that carries its value in reuse is the same component that carries its evidential force and its hazard, so no design separates them and the paper states a trade-off in place of a solution. And records of research process are themselves an object of downstream interest, which raises the question of terms under which they may be reused. The paper closes with the governance of the grounds, the detection of misapplication, the treatment of jurisdictional variation, and the conditions under which the account would be shown wrong.

Keywords. disclosure governance; information hazard; appropriability; research commons; open science

Draft. This is a working draft. It is circulated for comment, and its arguments, formulations, and open questions are subject to revision. Correction and criticism are welcome at huangwanhong@serendip.ngo.

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Use of generative artificial intelligence. Generative AI assistants (Anthropic’s Claude and OpenAI’s ChatGPT) were used extensively in preparing this manuscript: for literature survey, for drafting prose from the author’s specifications, for adversarial review of the argument, for the construction of the figures, for consistency and citation auditing, and for typesetting. Formulations and objections arose in the course of that exchange as well as from the author. Every claim, argument, and citation was reviewed and decided by the author, who is responsible for the content and for any errors. Bibliographic details are to be verified against the primary sources before reliance.

Suggested citation: Wanhong Huang, *Grounds of Restriction and the Governance of Disclosure in Generative Relational Academic Infrastructure*, working draft, 2026.

1 Introduction

This section states the question the paper treats, why the answer given by ownership is unavailable, the position taken, and the structure of the argument. Its role is to fix the undertaking. Its method is expository.

An arrangement that records how an understanding changed will hold material that cannot be disclosed to everyone. Some of it concerns instruments that admit limited use. Some of it would create danger if propagated. Some of it is early and would be damaged by exposure. Some of it funds the work through excludability.

The question is not whether such an arrangement may withhold, since plainly it must. The question is on what ground, and what follows from the ground.

The companion papers answer part of this by refusing an answer given by ownership. Nothing in the design confers a right to exclude, and attribution is deliberately modelled on the side of intellectual property that secures credit without conferring control. Withholding must therefore be justified by a reason rather than by a title, and the reason must do work: it must determine what is withheld, and it must determine what remains.

Three reasons, developed in the companion papers and restated here.

Legal instruments protect finished products (Boyle, 2008). Copyright attaches to fixed expression, patent to a completed invention with disclosure, trade secret to information whose value depends on its concealment. A party's position within an ongoing generation is protected by none of them, and the cumulative character of research makes the omission consequential (Scotchmer, 1991; Lemley, 2012), so an arrangement of this kind cannot borrow the ownership answer even if it wished to.

Rights to exclude, multiplied across many fine-grained contributions, produce the fragmentation documented in biomedical research, where downstream work requires so many permissions that it does not occur (Heller and Eisenberg, 1998).

And the design's own criterion runs the other way. An arrangement in which value converges on a party outside the relation that produced it is the condition the series diagnoses, and a scheme of exclusive rights over trajectory content would install exactly that.

The paper holds that restriction is legitimate, is unavoidable, and is governed by the reason for which it is imposed.

It further holds that a ground of restriction determines an object. Where the reason for withholding is that an instrument admits one user at a time, nothing about the record of what was done with the instrument follows. Where the reason is that a method creates danger when propagated, the method is what must be withheld and the questions and decisions surrounding it are not. Confusing the grounds is how a legitimate reason becomes a general licence.

And it holds that the residue matters more than the restriction. What a ground leaves disclosable is what distinguishes an arrangement that withholds for a reason from one that withholds because it can.

The paper proceeds in five parts.

The first fixes terms and the structure of a ground. The second treats the four grounds

in turn and then compares them, establishing that each restricts a different object and leaves a different residue, and that they are irreducible to one another.

The third treats three difficulties the grounds create rather than resolve: the displacement of the governance problem when exploration moves into simulation, the fact that value and hazard are carried by the same component of a record, and the downstream use of trajectory records by parties who did not produce them.

The fourth treats governance: who may restrict, how misapplication is detected, and how an arrangement operating across jurisdictions handles legal variation without encoding law.

The fifth states predictions and the conditions under which the account would be shown wrong.

2 Scope, Terms, and the Structure of a Ground

This section fixes the terms and the analytic instrument the paper uses. Its role is to make the four grounds comparable, by fixing in advance what a ground is required to state. Its structure treats the terms, the structure of a ground, the closure of the set, the requirement that a residue be disclosed, and the standard by which a restriction is assessed. Its method is stipulative, and the substance of the grounds is developed in the sections that follow.

2.1 Terms

Definition 2.1. (Disclosure class). *A disclosure class is a value attached to a record, governing the set of parties to whom it may be disclosed. Classes are ordered by inclusion. Their identity and membership are settled by the operating charter of a community and not by this account, in the manner of the collective-choice arrangements by which a group settles its own operational rules (Ostrom, 1990).*

Definition 2.2. (Ground). *A ground is the declared reason for which a record is disclosed at less than the widest class available. A ground is a property of a restriction and not of a record: the same content may be restricted on different grounds in different arrangements, and the consequences differ accordingly.*

Definition 2.3. (Residue). *The residue of a ground is what remains disclosable at the widest class when a restriction on that ground is in force. A ground with an empty residue licenses total silence.*

2.2 The Structure of a Ground

Requirement 2.4. (What a ground must state). *A ground is admissible in this account only where it states four things: the condition under which it obtains; the object it restricts; the residue it leaves disclosable; and the failure produced by applying it where its condition does not obtain.*

The fourth item is the one that makes the instrument useful. A ground stated without its failure mode is a licence, since any party may assert the condition and no party can say what has gone wrong when the assertion is false.

Each of the four sections that follow states its ground in this form, and the comparison assembles them.

2.3 The Closure of the Set

Requirement 2.5. (Closed set of grounds). *The set of grounds is closed. A community may not declare a new ground in its charter, and a restriction that fits none of the four is inadmissible.*

The reason is the one the requirement guards against. Grounds determine what may be withheld, so a community free to invent grounds is free to withhold anything by naming a reason, and the apparatus becomes decorative.

The cost is real and is stated. A restriction that is reasonable and fits none of the four grounds cannot be imposed within a conforming arrangement, and the account may therefore be wrong by omission. The falsification section treats this directly, since a reasonable restriction fitting no ground is precisely what would show the set incomplete.

2.4 The Residue Requirement

Requirement 2.6. (Disclosure of the residue). *Where a restriction is imposed on a ground, the residue corresponding to that ground is disclosed at the widest class available. A restriction withholding material that its ground does not cover is inadmissible, whatever the ground's condition.*

This is the operative requirement of the paper and the one on which everything later depends. Without it, every ground collapses into the same thing, which is silence with a justification attached, and the distinctions the following sections draw would have no consequence.

2.5 The Standard of Assessment

Requirement 2.7. (Assessment of a restriction). *A restriction is assessed by three questions: whether the ground's condition obtains; whether the object restricted is the object the ground identifies; and whether the residue has been disclosed. A restriction failing any of the three is misapplied, and the account names the corresponding failure.*

Two features of the standard are worth stating.

It is applied by readers and not by a protocol. The specification records the declared ground and enforces the class, and it cannot assess whether the condition obtains. The assessment is therefore a social act performed on a visible record.

And it is applied to a restriction and not to a party. A party may impose one restriction properly and another improperly, and the account draws no inference from the one to the other.

2.6 Summary of Results

This section has fixed the terms and the instrument. Definition 2.1 to 2.3 fix the disclosure class, the ground and the residue, with a ground a property of a restriction in place of a property of content. Requirement 2.4 requires a ground to state its condition, its object, its residue and the failure produced by misapplication, the last being what prevents a ground from operating as a licence. Requirement 2.5 closes the set of grounds, at the stated cost that a reasonable restriction fitting none of them cannot be imposed. Requirement 2.6 requires the residue to be disclosed and is the requirement on which the paper's distinctions depend. Requirement 2.7 fixes the three-question standard by which a restriction is assessed, applied by readers to restrictions and not by protocols to parties.

3 Resource Rivalry and the Selectivity of Allocation

This section treats the first ground. Its role is to establish that scarcity of an instrument justifies restricting access to the instrument and justifies nothing about the record of what was done with it, which is the confusion most often made in practice. Its structure states the ground in the form fixed in Section 2, then treats the allocation that scarcity forces, the proposal that follows, and the limits. Its method is analysis, with the allocation argument taken from the companion paper on incentives.

3.1 The Ground

Condition. A resource required for the work admits limited simultaneous use: an instrument, a facility, a quantity of compute, a clinical population, a budget line.

Object restricted. Access to the resource.

Residue. The trajectory in full. Questions, methods, decisions, failures, interpretations and results are all disclosable, since none of them consumes the resource.

Failure on misapplication. Where the record of work done with a scarce instrument is withheld on the ground of rivalry, a restriction on access has been converted into a restriction on knowledge, and the ground supplies no support for the conversion.

Claim 3.1. *Rivalry of a resource justifies selectivity in access to the resource and justifies no restriction on the record. A record is non-rival, and the condition that grounds the restriction does not obtain of it. This claim is formal.*

The claim is worth stating flatly because the conversion is routine. Laboratories with restricted instruments commonly restrict the surrounding record as well, and the practice has reasons, which are appropriability, hazard or exploratory vulnerability. Those are different grounds with different residues, and they must be declared as themselves.

3.2 The Allocation That Scarcity Forces

Scarcity of an instrument forces a selection, and the companion paper on incentives establishes what a selection does: it is a contest, its aggregate effort is dissipated, and what

distinguishes an arrangement that generates from one that merely selects is whether content from unselected work enters what proceeds with attribution.

Claim 3.2. *The rivalry ground bears on access and the selection it forces bears on effort, and the two are separate questions. An arrangement may allocate an instrument selectively and remain generative in the sense of the companion paper, provided the work of unsuccessful applicants is absorbed with attribution rather than discarded. This claim is formal.*

3.3 The Proposal That Follows

A proposal follows from the two claims together, and it is the paper's principal positive contribution on this ground.

Claim 3.3. *Where an instrument is scarce, the deciding process may be opened without opening the instrument. Proposals, the reasoning behind them, the objections raised, the decision and its stated reason are all within the residue of the rivalry ground, and disclosing them costs nothing in the resource. What is thereby opened is the direction of the work rather than the capability to perform it. This claim is formal.*

Two consequences.

The record of an allocation decision becomes inspectable, which bears on the misapplication failure of this section and on the governance treatment later.

And the unsuccessful proposals become available, which is the condition under which the absorption test can be satisfied at all. An arrangement whose declined proposals are unrecorded cannot absorb from them.

3.4 The Limits

Three.

Opening the deciding process does not reduce the dissipation of the contest. It converts a portion of the dissipated effort into a retained record, which is a different thing and is worth less than not dissipating it.

Opening the deciding process exposes the applicants. A declined proposal is a record of a party's judgement that did not persuade, and in a community that reads such records unfavourably the exposure is a cost falling on the unsuccessful. The exploratory-vulnerability ground bears on this and is treated in its own section.

And the claim that opening the process is costless in the resource is true of the instrument and false of the assessors, whose capacity the companion paper identifies as the binding constraint. A more open deciding process consumes more assessment.

Claim 3.4. *Opening the deciding process is costless in the rival resource and is not costless. It consumes assessment capacity, which is itself subtractable, and it exposes unsuccessful applicants. Both costs fall outside the rivalry ground and must be justified separately. This claim is formal.*

3.5 Summary of Results

This section has stated the rivalry ground: its condition is a resource admitting limited simultaneous use, its object is access to that resource, its residue is the trajectory in full, and its misapplication converts a restriction on access into a restriction on knowledge. Claim 3.1 states that the record is non-rival, so the ground's condition does not obtain of it, and that restrictions on the record commonly imposed under this heading are appropriability, hazard or vulnerability and must be declared as themselves. Claim 3.2 separates the ground from the selection scarcity forces. Claim 3.3 states the proposal that follows: the deciding process may be opened without opening the instrument, which discloses the direction of the work while withholding the capability. Claim 3.4 records that opening the process consumes assessment capacity and exposes unsuccessful applicants, costs which fall outside this ground.

4 Knowledge Hazard and the Boundaries of Trajectory Disclosure

This section treats the ground on which the design's central tension appears. Its role is to state the ground, to establish that restricting the instrument accomplishes nothing where the hazard lies in the method, and to identify the mechanism available where the ordinary answers are unavailable. Its structure states the ground, then treats the location of the hazard, the residue, the mechanism of conditional access, and the limits. Its method is analysis, with the hazard literature taken as given.

4.1 The Ground

Condition. Propagation of the content would create a risk of serious harm, and the risk arises from the content being known in place of from any use of a resource.

Object restricted. The propagable content of the method: the sequence by which a capability is obtained, the constraints discovered, the parameters that matter, the practical technique.

Residue. The existence of the work, the questions pursued, the decisions taken and their reasons, the interpretations reached, and results whose disclosure does not convey the method.

Failure on misapplication. Where hazard is declared of content whose propagation creates no risk, a restriction of the widest kind is obtained on the strongest available justification, and the declaration is unfalsifiable from outside the restricted class.

4.2 The Location of the Hazard

Claim 4.1. *Where the hazard lies in the propagation of a method, restricting access to an instrument accomplishes nothing. The instrument is frequently replaceable and often widely held, and a party who obtains the method may obtain an instrument elsewhere. The rivalry ground and the hazard ground therefore restrict different objects and are not substitutes. This claim is formal.*

The claim disposes of a design that would otherwise be attractive and that the source

discussion for this series proposed at length: restrict the instrument, open the reasoning. That arrangement is correct under rivalry and is exactly wrong under hazard, since the reasoning is where the hazard lives.

The typology of harms that can follow from information being known is developed in the literature on information hazards (Bostrom, 2011), and this account takes from it the point that the hazard attaches to knowing rather than to holding.

4.3 The Residue and Its Extent

The residue of the hazard ground is larger than practice usually allows, and the reason is worth stating.

That a line of work exists, what question it pursues, what obstacles it has met, what directions it has abandoned and why, and what its participants take their results to mean: none of these conveys a capability. A party learning that a group is pursuing a question, and that a particular approach failed for a stated reason, is not thereby closer to performing anything.

Claim 4.2. *The hazard ground leaves a substantial residue, and an arrangement withholding the existence of work, its questions, its decisions and its interpretations on the hazard ground has withheld material the ground does not cover. Under the residue requirement of Section 2 such a restriction is misapplied. This claim is formal.*

The practical consequence is the arrangement the companion papers describe for restricted work: the trajectory is visible in its structure and its reasoning, and the propagable method is held at a narrow class.

4.4 Conditional Access as the Available Mechanism

Where a party outside the restricted class needs the method, the mechanisms available are those developed for capability without distribution.

The pattern is that access is supplied under conditions rather than by transfer: a party uses a capability without receiving the artefact that confers it, and the conditions of use are enforced by the supplying party (Shevlane, 2022). Applied here, the corresponding arrangement is that a party is admitted to the restricted class under a charter's assurance requirements, with the admission and its ground recorded.

Claim 4.3. *The hazard ground admits no general solution and admits a graded one. Conditional access converts a binary question, whether to disclose, into a question about whom to admit and under what commitments, and the conversion is what makes work under this ground possible at all. This claim is formal.*

4.5 The Limits

Four, and the third is the one that constrains the whole series.

Assessment of hazard is not a competence a protocol has, and is frequently not a competence the participants have either. The account requires the ground to be declared

and cannot supply the judgement it declares, and the governance of research of concern proceeds by institutional review for the same reason (Bostrom, 2011).

Misapplication is undetectable from outside. A party outside the restricted class cannot assess whether the content restricted is in fact propagable, so the assessment standard of Section 2 is available only to parties within the class or to a body a charter empowers.

The hazard is bounded by what the record transmits. The companion papers establish that a record carries the articulable fraction of what its participants held, so a hostile reader obtains what was written and not the capability that was not. This bounds the hazard and bounds the benefit identically, which is treated in its own section.

And a restriction on this ground has no natural terminus. Rivalry ends when the instrument is free, appropriability ends when the funding purpose is served, vulnerability ends when the work matures. A hazard does not end, so scheduled release is unavailable here, and the class is maintained indefinitely by parties who must continue to exist.

Claim 4.4. *The hazard ground is the only one of the four for which no scheduled release is available, and a restriction imposed on it is therefore a permanent commitment by a community to maintain a class. Where the community ceases, the material is held by nobody under any charter, and the account offers no treatment of that case. This claim is formal.*

4.6 Summary of Results

This section has stated the hazard ground: its condition is that propagation creates risk, its object is the propagable content of a method, its residue is existence, questions, decisions, interpretations and non-conveying results, and its misapplication obtains the widest restriction on the strongest justification without being falsifiable from outside. Claim 4.1 establishes that restricting an instrument accomplishes nothing where the hazard lies in the method, which disposes of the restrict-the-instrument-and-open-the-reasoning design. Claim 4.2 establishes that the residue is substantial and that withholding it is misapplication. Claim 4.3 identifies conditional access as the graded mechanism that makes work under this ground possible. Four limits are recorded, of which the fourth is distinctive: hazard admits no scheduled release, so a restriction on this ground commits a community to maintaining a class indefinitely, and the account has no treatment of the case in which the community ceases.

5 Exploratory Vulnerability, the Timing of Exposure, and the Observer Effect

This section treats the ground that concerns early work. Its role is to establish that the ground restricts timing rather than content, to state the asymmetry that prevents it from becoming permanent withholding, and to treat the effect of recording on what is recorded. Its structure states the ground, then treats the monotonicity of disclosure, the observer effect, the two populations affected differently, and the limits. Its method is analysis.

5.1 The Ground

Condition. Exposure of the work at its present stage would suppress or distort it: through premature criticism of an idea that is not yet formulated, through appropriation by a better-positioned party, or through the participant's own reluctance to record what may be read.

Object restricted. The timing of exposure.

Residue. Everything, at the scheduled time. The ground justifies delay and justifies no permanent withholding.

Failure on misapplication. Where a delay is extended indefinitely, or where a schedule is set without a terminus, a permanent withholding has been obtained under a ground that licenses only postponement.

5.2 The Monotonicity of Disclosure

Claim 5.1. *Disclosure is monotone: it may widen and cannot narrow, since a party who has read a record retains what they read. An arrangement offering the appearance of withdrawal misdescribes to its own participants a state of affairs obtaining outside it. This claim is formal.*

Two consequences for this ground.

The default must be restrictive, since an error in the direction of openness is uncorrectable and defaults are where such errors occur.

And a schedule may be shortened and must not be extended. The asymmetry is the substance of the ground: a delay that can be extended indefinitely is a permanent withholding that has been made to look temporary, and the misapplication failure above is exactly its instrument.

5.3 The Observer Effect

A difficulty specific to this ground is that recording alters what is recorded.

Parties speak differently when a record is being made. The material this series most wants preserved, half-formed ideas, real disagreement, the remark someone makes before they are sure of it, is the material least likely to be produced under observation.

Claim 5.2. *An arrangement that records deliberation can degrade the deliberation it records, and the degradation falls on the content the arrangement most values. The vulnerability ground therefore concerns the conditions of production and not only the safety of the producer. This claim is a conjecture, and the mechanism it posits is widely reported and not, so far as this account is aware, measured for research settings.*

The mitigations available are all partial and are stated as such: recordings default to the narrowest class, release is scheduled rather than immediate, and participants may mark passages as excluded from the record. None removes the effect, and an arrangement claiming otherwise is claiming that people behave identically observed and unobserved.

The design's structural mitigation is the unit of record. What the arrangement preserves is that an objection altered an assumption, and not that a particular party expressed it clumsily at a particular moment, which reduces the exposure without eliminating the awareness of being recorded.

5.4 The Two Populations

The ground operates differently on two populations and the difference determines how a charter should set it.

A well-positioned party is exposed to criticism and not to appropriation, since standing protects the association of an idea with its author (Merton, 1957, 1968). Delay serves them weakly.

A party without position is exposed to both, and appropriation is the exposure that matters. Delay serves them strongly, and the registration mechanism of the companion specification serves them further, since it separates the making of a claim from its disclosure.

Claim 5.3. *The vulnerability ground protects parties without standing more than parties with it, and a charter setting short delays uniformly imposes the exposure on the population least able to bear it. This claim is formal, given the asymmetry in how standing protects the association of an idea with its author.*

5.5 The Limits

Three.

Delay costs the work the objections it would have received. A participant who withholds an idea until it is safe has protected it from appropriation and has also protected it from improvement, and the arrangement has no way of supplying the second without the first.

The ground has no objective terminus. Whether work has matured enough to expose is a judgement, and a party may in good faith judge indefinitely that it has not.

And the ground is the easiest of the four to assert. Its condition concerns the state of a piece of work, which only its participants can assess, and its misapplication is therefore detectable only from the pattern of a party's schedules over time.

5.6 Summary of Results

This section has stated the vulnerability ground: its condition is that exposure at the present stage would suppress or distort the work, its object is the timing of exposure, its residue is everything at the scheduled time, and its misapplication converts postponement into permanent withholding. Claim 5.1 fixes monotonicity, from which follow a restrictive default and the asymmetry that a schedule may be shortened and not extended. Claim 5.2 records the observer effect, which degrades exactly the content the arrangement most values, with partial mitigations and the structural mitigation of the recording unit. Claim 5.3 establishes that the ground protects parties without standing

more than parties with it, so uniform short delays impose exposure on those least able to bear it. Three limits: delay forgoes the objections that would have improved the work, the ground has no objective terminus, and it is the easiest of the four to assert, its misapplication being detectable only from a pattern over time.

6 Appropriability and the Funding of Restricted Trajectories

This section treats the ground on which most restriction in practice actually rests, and which is most often declared as something else. Its role is to state it plainly, to establish that it is legitimate within limits, and to identify the failure it invites. Its structure states the ground, then treats the legitimacy of the condition, the residue, the relation to the incumbent instruments, and the misapplication that gives the ground its bad name. Its method is analysis.

6.1 The Ground

Condition. The work is funded by an arrangement that requires excludability: a commercial party recovering an investment, a licensing arrangement, a competitive advantage that pays for the laboratory, a dataset whose collection was financed by restricting access to it.

Object restricted. The content whose disclosure would destroy the excludability on which the funding depends.

Residue. The existence of the work, the questions pursued, the decisions taken and their reasons, and negative results, none of which conveys the excludable content.

Failure on misapplication. Where appropriability is declared and nothing is in fact appropriated, a general secrecy has been obtained on an economic justification that does not apply, and the failure has a name: rent-seeking secrecy.

6.2 The Legitimacy of the Condition

The condition is legitimate and this account does not treat it as a regrettable concession.

Some work is not performed unless somebody pays for it, and some funding arrangements require that the funder be able to exclude. Where that obtains, an insistence on disclosure does not produce a disclosed trajectory; it produces no trajectory, since the work does not occur.

Claim 6.1. *The appropriability ground is a condition of some work existing at all, and an account of restriction that treated it as illegitimate would be an account under which the work it wishes to record is not performed. This claim is formal.*

The economics of disclosure supplies the general form: where the value of information cannot be assessed without disclosing it, and disclosure destroys the ability to charge for it, private provision requires an instrument of exclusion (Arrow, 1962).

6.3 The Residue

The residue of this ground is the same in shape as that of hazard, and the reason differs.

Under hazard, what may be disclosed is what does not convey a capability. Under appropriability, what may be disclosed is what does not destroy excludability, which is a narrower test in some cases and a wider one in others. A negative result may be commercially harmless and hazardous; a question may be commercially revealing and entirely safe.

Claim 6.2. *The residues of hazard and appropriability differ in content although they coincide in shape, and a restriction declaring one while applying the residue of the other is misapplied. This claim is formal.*

Two items are within the residue in almost all cases and are worth naming, since practice withholds them routinely.

The existence of the work. That a party is pursuing a question is rarely the content whose disclosure destroys excludability, and its disclosure is what permits others to avoid duplicating the effort.

Negative results. An approach that failed is rarely what a competitor would have paid for, and its disclosure is the principal contribution this series claims for the reuse of abandoned material.

6.4 The Relation to the Incumbent Instruments

The ground overlaps the existing legal apparatus and is not identical to it.

Trade secret protects information whose value depends on its concealment (Boyle, 2008), and is the instrument that corresponds most closely. Patent protects an invention in exchange for disclosure, so a party who patents has already accepted disclosure of the claimed matter and the ground does not extend to it. Copyright is irrelevant here, since it protects expression and not the content whose excludability is at issue.

Claim 6.3. *A party holding a patent has exchanged disclosure for exclusion and may not invoke appropriability over the disclosed matter. A party holding a trade secret has an instrument whose operation the ground describes. The account therefore adds nothing to what the law permits and constrains what a party may declare within a conforming arrangement. This claim is formal.*

6.5 Rent-Seeking Secrecy

The misapplication of this ground is common enough to deserve its own treatment.

A restriction declared on appropriability where no excludable value exists obtains secrecy at no economic cost to the declaring party and at a cost to everyone else. It is attractive precisely because the condition is unverifiable from outside: whether disclosure would destroy a funding arrangement is a matter of the declaring party's commercial position, which they alone know.

Claim 6.4. *Appropriability is unverifiable from outside the declaring party, and its misapplication is therefore detectable only through the residue. A party declaring appropriability and disclosing the residue has made a claim that costs them something; a party declaring it and withholding the residue has made a claim that costs them nothing, and the second is the observable signature of the misapplication. This claim is formal.*

The observation is the strongest argument in this paper for the residue requirement. Where residues are disclosed, a false declaration is expensive; where they are not, a false declaration is free.

6.6 The Limits

Three.

The ground has a terminus in principle and rarely in practice. Excludability is required until the funding purpose is served, and no mechanism obliges a party to release when it has been.

The ground is available to parties who have funding arrangements and unavailable to those who do not, so it is asymmetric in a direction that favours the well-resourced, which is the direction this series generally resists and cannot resist here.

And the account offers no procedure for assessing a declaration. The three-question standard of Section 2 reduces here to one question a reader can answer, which is whether the residue was disclosed.

6.7 Summary of Results

This section has stated the appropriability ground: its condition is that funding requires excludability, its object is the content whose disclosure would destroy it, its residue is existence, questions, decisions and negative results, and its misapplication is rent-seeking secrecy. Claim 6.1 establishes the condition's legitimacy, since work under it is not performed otherwise. Claim 6.2 distinguishes the residues of hazard and appropriability, which coincide in shape and differ in content. Claim 6.3 relates the ground to trade secret and excludes matter already disclosed under patent. Claim 6.4 identifies the observable signature of misapplication: a declaration accompanied by the residue costs the declaring party something, and a declaration withholding it costs nothing, which is the strongest argument for the residue requirement. Three limits: the terminus is unenforced, the ground favours the well-resourced, and assessment reduces to the residue question.

7 The Four Grounds Compared and the Operation of the Residue Rule

This section assembles the four grounds and states what follows from having four rather than one. Its role is to establish irreducibility, to show how grounds compose where several obtain, and to state the rule that governs the composition. Its structure treats the comparison, irreducibility, composition, the ordering of restrictions, and what the apparatus achieves. Its method is comparison, with the table as the instrument.

Table 7.1: The four grounds: condition, object restricted, residue, and the failure produced by misapplication

Ground	Condition	Object restricted	Residue and failure
Rivalry	A resource admits limited simultaneous use	Access to the resource	The trajectory in full. Failure: a restriction on access is converted into a restriction on knowledge
Hazard	Propagation of the content creates risk	The propagable content of the method	Existence, questions, decisions, interpretations, non-conveying results. Failure: the widest restriction on the strongest justification, unfalsifiable from outside
Vulnerability	Exposure at this stage suppresses or distorts the work	The timing of exposure	Everything, at the scheduled time. Failure: postponement converted into permanent withholding
Appropriability	Funding requires excludability	Content whose disclosure destroys excludability	Existence, questions, decisions, negative results. Failure: rent-seeking secrecy

7.1 The Comparison

Table 7.1 assembles the four grounds in the form fixed in Section 2.

7.2 Irreducibility

Claim 7.1. *No ground reduces to another. Rivalry restricts a resource and the others restrict content. Hazard restricts what a reader could do and appropriability restricts what a competitor could sell, and the two select different content from the same trajectory. Vulnerability restricts a time and the others restrict a set of parties. Each pair differs in the object restricted, so a single ground could not do the work of the set. This claim is formal.*

Two consequences.

A restriction must declare which ground it invokes, since the object and the residue follow from the ground and not from the content.

And an arrangement using one ground for everything has either misapplied it or is operating under a condition that does not obtain. The most common instance is an arrangement declaring commercial sensitivity over material that is in fact restricted because it is dangerous, or because it is early, or because the instrument is booked.

7.3 Composition

Several grounds frequently obtain at once. Pharmaceutical development is funded by excludability, produces material whose propagation carries risk, and involves early work that exposure would distort.

Requirement 7.2. (Composition of grounds). *Where several grounds obtain, each is declared separately, and the residue disclosed is the intersection of the residues of the grounds declared. A party may not declare one ground and withhold on the strength of another.*

The intersection rule is what prevents composition from becoming a route around the residue requirement. A party declaring all four grounds obtains the smallest residue the four permit, and must be prepared to state, for each, that its condition obtains.

Claim 7.3. *Composition narrows the residue and raises the number of conditions a party asserts. Declaring more grounds is therefore not free, since each declaration is an assertion that can be assessed and that a reader may find false. This claim is formal.*

7.4 The Ordering of Restrictions and Their Release

The grounds differ in how they end, and the difference determines what a charter must arrange.

Vulnerability ends on a schedule, which the arrangement records and which may be shortened.

Rivalry ends when the resource is no longer contended, which is observable and which no mechanism enforces.

Appropriability ends when the funding purpose is served, which is known to the declaring party alone.

Hazard does not end.

Claim 7.4. *Of the four grounds only vulnerability admits a recorded terminus, so an arrangement in which most restrictions are declared on the other three accumulates restricted material indefinitely. The accumulation is a property of the grounds and not of any implementation, and a community that wishes its record to open over time must prefer the ground that ends. This claim is formal.*

7.5 The Achievements of the Apparatus

Three things, stated modestly.

It makes a restriction an assertion. A party restricting a record asserts a condition, and the assertion is recorded, dated and attributable, where under present practice a restriction is a default that asserts nothing.

It makes misapplication describable. Each ground has a named failure, so a reader who believes a restriction improper has a vocabulary in which to say what is wrong with it.

And it makes the residue the object of attention. The question a reader asks is not whether something was withheld, which is uninformative, but whether what was disclosed matches the ground declared.

Claim 7.5. *The apparatus supplies a vocabulary and a checkable question and supplies no enforcement. Whether restrictions are properly declared depends on whether communities attend to the residues, and the account offers no mechanism that makes them attend. This claim is formal.*

7.6 Summary of Results

This section has assembled the four grounds in Table 7.1 and established their irreducibility, since each pair differs in the object restricted. Two consequences follow: a restriction must declare its ground, and an arrangement using one ground for everything has misapplied it. Requirement 7.2 fixes composition by intersection of residues, which prevents composition from evading the residue requirement and makes each additional declaration an assertion that may be found false. Claim 7.4 establishes that only vulnerability admits a recorded terminus, so an arrangement resting on the other three accumulates restricted material indefinitely, and a community wishing its record to open must prefer the ground that ends. Claim 7.5 states what the apparatus achieves: a restriction becomes a dated attributable assertion, misapplication becomes describable, and the residue becomes the object of attention, with no enforcement supplied.

8 Simulation Environments and the Displacement of the Governance Problem

This section treats the most attractive proposal for escaping the rivalry ground, and establishes that it relocates the difficulty in place of removing it. Its role is to prevent a design that appears to dissolve scarcity from being adopted without its consequences understood. Its structure treats the proposal, the displacement, the benchmark as an attractor, the validation dependence, and the effect on hazard. Its method is analysis, and the section reaches a negative result with one qualification.

8.1 The Proposal

Where exploration can be conducted in a simulated environment, the instrument is required only at the end, to test what the simulation has selected. Many parties may then explore, the scarce resource is consumed once, and the rivalry ground applies to a narrow final stage in place of the whole undertaking.

The proposal is genuine and has instances. Community simulation followed by shared fabrication runs in hardware design; open computational exploration precedes experimental validation in several fields.

Claim 8.1. *Where a simulated environment is faithful enough for exploration, the rivalry ground contracts to the validation stage, and the earlier stages become open to any party with computation. This claim is formal, conditional on fidelity, which is a matter for each field.*

8.2 The Displacement

What the proposal does not do is remove the governance question, and the reason is structural.

Claim 8.2. *A simulated environment is a selection environment. What it admits, what it rewards, what it renders visible and what it cannot represent determine which explorations succeed, so the party who specifies the model and the benchmark occupies the position that the party allocating the instrument previously occupied. The governance problem moves from the allocation of a resource to the specification of an environment and is not reduced. This claim is formal.*

The displacement is unfavourable in one respect. Allocation of an instrument is visible, episodic and contestable: a decision is made, a party is refused, and the refusal can be challenged. Specification of a benchmark is continuous, technical and rarely contested, and its effects appear as properties of the field rather than as decisions anyone took.

8.3 The Benchmark as an Attractor

The natural remedy is to open the model and the benchmark, and the remedy is insufficient for a reason the companion papers develop.

Claim 8.3. *An open benchmark is a public attractor. Parties optimise for what it measures, the categories it recognises become the categories the field maintains, and inspectability of its assumptions does not prevent this. Openness of a selection environment makes its criteria visible and does not prevent them from governing. This claim is formal, and the mechanism is the reactivity documented for public measures (Espeland and Sauder, 2007).*

The consequence for the design is that a simulated exploration stage is subject to the sparsity and plurality analysis of the companion paper on incentives. A single widely adopted benchmark, applied continuously, is the condition under which reactivity is strongest, and it is also the arrangement that a field finds most convenient.

8.4 The Validation Dependence

A second consequence is easily overlooked.

A simulation's fidelity is established against the physical apparatus, so the open layer inherits a dependence on the closed one. The parties who validate the model hold a position of the kind the companion papers govern, and the validation itself consumes the scarce resource whose contraction the proposal claimed.

Claim 8.4. *The open exploration layer depends on validation performed with the scarce resource, so the resource is consumed by the arrangement that was meant to economise on it, and the parties performing validation acquire a position over what the open layer may take as fidelity. This claim is formal.*

8.5 The Effect on Hazard

The proposal's effect on the hazard ground runs in both directions and the direction depends on where the hazard lies.

Where the hazard lies in physical execution, simulation lowers it: exploration produces no dangerous artefact and the dangerous stage is confined and supervised.

Where the hazard lies in design, simulation raises it. An environment that lets many parties explore a design space cheaply is an environment that lowers the cost of arriving at a dangerous design, and the biological case is the one usually named.

Claim 8.5. *Simulation lowers hazard where the hazard lies in execution and raises it where the hazard lies in design. A proposal to displace exploration into simulation must therefore state which kind of hazard the domain carries, and the displacement is not neutral with respect to the hazard ground. This claim is formal.*

8.6 The Qualification

One thing the proposal achieves and the section should not obscure.

It changes who may participate in the exploration, from parties with access to an instrument to parties with computation, and the second population is far larger and differently distributed. This is a real widening and is the strongest reason to pursue the proposal, notwithstanding everything above.

Claim 8.6. *Displacement into simulation widens participation in exploration and relocates rather than reduces the governance problem. Both are true and the second does not defeat the first. This claim is formal.*

8.7 Summary of Results

This section has treated the simulation proposal. Claim 8.1 grants that the rivalry ground contracts to the validation stage where fidelity permits. Claim 8.2 establishes the displacement: a simulated environment is a selection environment, and whoever specifies the model and benchmark occupies the position the allocator previously held, with the displacement unfavourable because specification is continuous, technical and rarely contested where allocation is visible and challengeable. Claim 8.3 establishes that opening the benchmark does not prevent it from governing, since an open benchmark is a public attractor. Claim 8.4 records the validation dependence, under which the open layer depends on the closed one and the scarce resource is consumed after all. Claim 8.5 establishes that simulation lowers hazard of execution and raises hazard of design. Claim 8.6 records the qualification: participation in exploration is genuinely widened, and this is not defeated by the displacement.

9 The Shared Component of Value, Evidence, and Hazard

This section states the result that governs every restriction in this account. Its role is to show that the material a restriction removes is the material that made the record worth having, so that no design achieves both openness and safety by selecting carefully. Its structure states the component, the three properties it carries, the consequence for design, the bound that limits all three together, and what remains available. Its method is argument from the companion papers' results, assembled here for the first time.

9.1 The Component

Definition 9.1. (The omitted component). *The omitted component of a trajectory is what a finished artefact does not contain: the approaches attempted and abandoned, the reasons for abandoning them, the constraints discovered, the parameters that mattered, the practical technique by which a result was obtained.*

9.2 The Three Properties

Proposition 9.2. *The omitted component carries three properties at once. It is what gives a trajectory record its value in reuse, since what transfers to another party's problem is the record of what was tried and what obstruction was met. It is what gives the record its evidential force, since a record of a completed result is cheap to fabricate while a record of the path is costly in proportion to the parties who attested it. And it is what carries the hazard, since a propagable method is a sequence of steps and constraints rather than a statement of a result. This claim is formal.*

The proposition assembles results the companion papers establish separately. What is new here is the observation that they concern one object.

9.3 The Consequence for Design

Claim 9.3. *No design achieves the epistemic benefit of an open trajectory while withholding the parts that carry risk, since the parts are the same parts. A proposal to publish the trajectory minus its dangerous content is a proposal to publish the record minus what makes it worth publishing. This claim is formal.*

The claim disposes of a class of arrangements that appear reasonable and are incoherent. It also explains why the account of this paper is a typology of grounds and residues in place of a filtering rule: the design cannot separate the valuable from the dangerous within the component, so it separates the parties to whom the component is disclosed.

Three consequences follow for the grounds already stated.

Under hazard, the residue is what lies outside the omitted component: existence, questions, decisions, interpretations. The residue is real and is not the part that carries the reuse value, so a hazard restriction genuinely costs the commons something.

Under appropriability, the same holds with a different boundary, and negative results usually fall outside the excludable content while remaining inside the omitted component, which is why they are the residue item this series presses hardest.

And under vulnerability, the component is disclosed later rather than never, which is why that ground costs the commons least.

9.4 The Bound That Limits All Three

The properties are bounded together as well as carried together.

A record transmits the articulable fraction of what its participants held, and the tacit remainder is transmitted, if at all, by working alongside them (Collins, 1974, 2010; MacKenzie and Spinardi, 1995). So a reader receives what was written and not the capability that was not, and this is true of a benign reader and a hostile one alike.

Claim 9.4. *The value in reuse, the evidential force and the hazard of a trajectory scale together with the articulable fraction. A domain in which little of what matters can be written produces records of little use, little evidential weight and little danger; a domain in which much can be written produces records of much use, much weight and much danger. No design alters the ratio between them. This claim is formal.*

The claim has a practical use for a community setting its charter. Where participants find that little of what matters can be recorded, the arrangement will deliver little, and its restrictions may be correspondingly relaxed. Where much can be recorded, the arrangement delivers more and the grounds matter more.

9.5 The Mechanisms That Remain Available

The result is negative and leaves three things.

Disclosure to a restricted class preserves the component for the parties within it, which is what conditional access achieves and why it is the mechanism the hazard section identifies.

Delay preserves the component and postpones its availability, which is why vulnerability is the ground whose cost to the commons is temporary.

And the residues, being outside the component, are disclosable without touching the trade-off at all. This is the reason the residue requirement is the paper's operative rule: it is the only part of the design that gains something without giving something up.

Claim 9.5. *The residues are the part of the arrangement that costs nothing to disclose under any ground, and the trade-off between value and hazard applies only within the omitted component. An account that attended only to the component would find no room to move; the residue is where the room is. This claim is formal.*

9.6 Summary of Results

This section has stated the governing result. Definition 9.1 fixes the omitted component, and Proposition 9.2 establishes that it carries the reuse value, the evidential force and

the hazard of a trajectory at once. Claim 9.3 draws the consequence: no design publishes the trajectory minus its dangerous parts and retains the benefit, which is why this account is a typology of grounds in place of a filtering rule, and why it separates parties rather than content. Claim 9.4 establishes that all three properties are bounded together by the articulable fraction, so domains differ in how much any arrangement can deliver and endanger, and no design alters the ratio. Claim 9.5 records what remains: conditional access preserves the component for a class, delay postpones it, and the residues lie outside the component and are the only part of the design that gains without cost.

10 Downstream Use of Trajectory Records and the Terms of Their Reuse

This section treats a party the account has not so far considered: one who wants the records themselves, for purposes unrelated to the work they document. Its role is to establish that such a party exists, that the grounds so far stated do not reach them, and what a charter must therefore settle. Its structure treats the parties, the interest, why the existing grounds do not apply, the terms a charter must fix, and the limits. Its method is analysis, and the section reaches a requirement rather than a result.

10.1 The Parties and the Interest

A record of how understanding changed, at scale, across many trajectories, is an object of interest to parties who did not produce it and do not work on its subject matter.

It is training material (Coalition for Content Provenance and Authenticity, 2025), since a corpus of recorded reasoning with its transitions typed and its outcomes attached is precisely what a system learning to model reasoning would want. It is research material for the study of science itself. It is evaluation material for institutions assessing the parties who produced it. And it is commercial material for services built over it.

Claim 10.1. *The interest of downstream parties in trajectory records is proportional to the qualities the series claims for them. A record that is typed, attested, linked and complete is more valuable to a party who wants a corpus than an unstructured archive would be, so the design's success creates the interest. This claim is formal.*

10.2 The Inapplicability of the Existing Grounds

None of the four grounds reaches this case, and the reason is that all four concern the disclosure of content while this concerns its subsequent use.

Rivalry concerns a resource. Hazard concerns what a reader could do with a method. Vulnerability concerns exposure at a stage. Appropriability concerns the excludability that funds the work.

A party who obtains records lawfully disclosed at the widest class, and uses them to train a system or to build a service, has done nothing any ground addresses. The content was disclosed; the question is what may then be done with it.

Claim 10.2. *The four grounds govern to whom a record is disclosed and do not govern what a party may do with a record disclosed to them. The account is therefore incomplete with respect*

to downstream use, and the incompleteness is structural in place of an oversight. This claim is formal.

10.3 The Distributional Question

The question this raises is one the companion paper on incentives is equipped to state.

The labour of producing the record is performed by participants who bear the overhead, and the value of a corpus assembled from many such records accrues to whoever assembles it. Where the assembling party is a well-resourced third party and the producing parties are the population the series concerns, the arrangement transfers value from those with least to those with most, which is the pattern the folding criterion of the companion papers exists to identify.

Claim 10.3. *Uncompensated assembly of trajectory corpora by third parties is a case in which the value generated by a relation accrues outside it. The design's own criterion identifies the arrangement as extractive, and the design supplies no mechanism that prevents it. This claim is formal.*

10.4 The Matters a Charter Must Settle

The account's response is a requirement on charters in place of a fifth ground, since the matter concerns terms of use rather than conditions of disclosure.

Requirement 10.4. (Terms of downstream use). *A charter must state the terms on which records disclosed under it may be reused by parties other than their producers, and must state separately whether consent to disclosure constitutes consent to inclusion in a corpus assembled for purposes unrelated to the work. A record must carry the identifier of the charter under which it was disclosed, so that a party obtaining it can determine the terms.*

Three matters the requirement obliges a charter to decide and does not decide for it.

Whether reuse for study of the work's own subject matter differs from reuse for training a system.

Whether participants may withhold consent to corpus inclusion while consenting to open disclosure, and what a record looks like where some participants have and some have not.

And whether a licence is attached to the record, and if so which, given that licences designed for finished works apply awkwardly to a graph of attributed transitions.

10.5 The Limits

Three, and the first defeats most of what the requirement attempts.

Terms attached to a disclosed record are difficult to enforce and the difficulty is well known. A party who obtains records at the widest class and disregards the terms is reached only by legal action, which the producing parties are least placed to bring.

Consent to corpus inclusion is not meaningfully separable in practice. A record disclosed openly is available, and a term forbidding one use of available material is a term whose breach is undetectable.

And an arrangement that responded by restricting disclosure would defeat itself. Withholding records to prevent their assembly is a restriction on a ground the account does not admit, and it would cost the commons everything the openness was for.

Claim 10.5. *The terms requirement obliges a charter to make a decision and supplies no means of enforcing it, and the account records the difficulty as unaddressed. Restricting disclosure to prevent downstream assembly is not available, since it would impose the cost of the restriction on the commons in order to withhold a benefit from a third party. This claim is formal.*

10.6 Summary of Results

This section has treated downstream use. Claim 10.1 establishes that the interest of third parties in trajectory records is proportional to the qualities the series claims for them, so success creates the interest. Claim 10.2 establishes that the four grounds govern to whom a record is disclosed and not what may be done with it, so the account is structurally incomplete here. Claim 10.3 identifies uncompensated corpus assembly as a case in which value generated by a relation accrues outside it, which the design's own criterion marks as extractive and which the design does not prevent. Requirement 10.4 obliges a charter to state terms of reuse, to address corpus inclusion separately from disclosure, and to be identifiable from the record. Three limits are recorded: terms are hard to enforce, consent to corpus inclusion is not separable in practice, and restricting disclosure to prevent assembly would impose the cost on the commons to withhold a benefit from a third party.

11 Custodianship, Charters, and the Authority to Restrict

This section treats who may impose a restriction. Its role is to establish that the authority to restrict is itself a position, and to state the conditions under which it is admissible. Its structure treats the authority, the position it constitutes, the requirement of recorded exercise, the bound supplied by portability, and the custodian's own case. Its method is application of the companion papers' analysis of positions to the particular position this paper creates.

11.1 The Authority

Someone decides that a record is restricted, on what ground, at what class, and until when. The decision is not made by the protocol, which enforces a class it does not interpret, and it is not made by the law, which supplies constraints and no determinations.

Claim 11.1. *The authority to restrict is a position in the sense of the companion papers: it is constituted by acts the arrangement requires, its holder determines what other participants may see, and the capacity of others to act depends on its exercise. Every analysis the series applies to positions applies to it. This claim is formal.*

The claim has a consequence the section develops: an arrangement designed to resist concentration cannot exempt from that resistance the one authority that determines what participants know.

11.2 The Conditions of Admissibility

The companion papers hold that a position is admissible where the acts constituting it are recorded and attributable, and where participants retain the capacity to proceed without the holder's permission. Applied here the two conditions have specific content.

Requirement 11.2. (Recorded exercise). *Every imposition, alteration and release of a restriction is recorded as an operation carrying the party who performed it, the ground declared, the class imposed, the schedule where one applies, and the time. An arrangement in which a restriction can be imposed or altered without a record does not satisfy the first condition of admissibility.*

The requirement is cheap and is the whole of what the design can do about the authority. It does not prevent an improper restriction, and it makes the pattern of a party's restrictions inspectable, which is the only basis on which the misapplication of Section 6 can be detected at all.

Requirement 11.3. (Separation of the restricting authority from the recording authority). *The party who registers transitions and the party who determines disclosure classes should be distinguishable in the record, and a charter concentrating both in one party must state that it does so. An arrangement in which one party both registers and classifies holds the record and its visibility together, and the record then evidences nothing about the party who produced it.*

11.3 The Bound

The second condition of admissibility is supplied here as elsewhere by portability, and its operation in this case is worth stating precisely.

A participant subject to restrictions they consider improper may obtain the record they can see and continue elsewhere. What they cannot obtain is the material they were never shown, so portability bounds the restricting authority weakly: it permits departure with what one has and not with what one was denied.

Claim 11.4. *Portability bounds the restricting authority less than it bounds the other positions the series treats, since the object of the restriction is precisely what a departing party cannot take. The bound on this authority is therefore charter and community in place of exit, and the series' general answer is weakest at the position where the stakes are highest. This claim is formal.*

11.4 The Charter's Role

What remains is the charter, and the requirement on it follows from the preceding claim.

Requirement 11.5. (Charter statement of restricting authority). *A charter must state which parties may impose a restriction, on which grounds, at which classes, and whether any party may impose a restriction without a schedule. A charter that leaves the authority unstated has left it with whoever operates the implementation.*

The final clause is the observation the requirement exists for. Authority not allocated is not absent; it is held by whoever is in a position to exercise it, which is the operator, and the operator is the party whose interests the series has least reason to trust.

11.5 The Custodian's Own Case

The party maintaining an arrangement and the party operating an implementation of it face the same question, and the companion papers require it to be answered.

A custodian who both maintains a charter and operates the implementation determines what may be restricted and performs the restricting. The series' position is that the two functions are separated, or that the conflict is declared with the arrangements limiting it recorded, and that the specification is licensed so that a community may fork it and continue elsewhere, the availability of exit being what conditions the terms on which voice operates (Hirschman, 1970).

Claim 11.6. *The requirements this section imposes apply to the party proposing them. An arrangement in which the proposing party maintained the charter, operated the implementation and held the restricting authority would be an instance of the concentration the series diagnoses, and the declaration of that conflict is a condition of the account's coherence rather than a courtesy. This claim is formal.*

11.6 Summary of Results

This section has treated the authority to restrict. Claim 11.1 establishes that it is a position in the series' sense, so every analysis applied to positions applies to it. Requirement 11.2 fixes recorded exercise, which does not prevent improper restriction and makes a party's pattern inspectable, and Requirement 11.3 asks that the restricting and registering authorities be distinguishable, since a party holding both holds the record and its visibility together. Claim 11.4 establishes that portability bounds this authority weakly, because a departing party cannot take what they were never shown, so the series' general answer is weakest where the stakes are highest. Requirement 11.5 obliges a charter to allocate the authority explicitly, on the ground that unallocated authority is held by the operator. Claim 11.6 applies all of this to the party proposing it.

12 Misapplied Grounds and the Detection of Improper Restriction

This section treats what can be done when a ground is declared falsely. Its role is to establish what is detectable, by whom, and what follows from detection. Its structure treats the four failures, the asymmetry of detectability, the residue as the observable, the patterns visible over time, and the limits of detection without remedy. Its method is analysis, and the section reaches a modest result.

12.1 The Four Failures

Each ground has a named failure, stated in its own section and assembled here.

Under rivalry, a restriction on access is converted into a restriction on knowledge.

Under hazard, the widest restriction is obtained on the strongest justification, and the justification cannot be assessed from outside the restricted class.

Under vulnerability, postponement is converted into permanent withholding, ordinarily by extending a schedule or by never setting one.

Under appropriability, secrecy is obtained on an economic justification that does not apply, which is rent-seeking secrecy.

12.2 The Asymmetry of Detectability

The four differ markedly in how far an outside party can detect them, and the difference determines where attention should be spent.

Rivalry misapplication is the most detectable. The ground's residue is the trajectory in full, so any withholding of the record under a declared rivalry ground is visible on its face.

Vulnerability misapplication is detectable over time. A schedule that is extended, or a restriction imposed without a schedule, is visible in the record, and a party's pattern across many restrictions is more visible still.

Appropriability misapplication is detectable only through the residue, since the condition is a fact about the declaring party's commercial position.

Hazard misapplication is the least detectable. The condition cannot be assessed without the content, and the content is what is withheld.

Claim 12.1. *Detectability of misapplication is inversely related to the width of the restriction the ground supports. The ground supporting the widest restriction is the one whose misapplication is hardest to detect, and the ordering is a property of the grounds rather than of any arrangement. This claim is formal.*

The claim identifies where a charter's attention is required. Hazard restrictions cannot be policed by readers, so a community relying on them must place the assessment with a body it constitutes for the purpose, and must accept that the body then holds a position of the kind the previous section treats.

12.3 The Residue as the Observable

The general instrument is the one the paper has developed throughout.

Claim 12.2. *For every ground, the observable available to a reader is whether the residue was disclosed. A declaration accompanied by its residue is a claim that costs the declaring party something; a declaration withholding the residue is a claim that costs nothing, and the second is the signature that is available in every case, including the case of hazard where the condition itself cannot be assessed. This claim is formal.*

The claim is worth emphasis because it turns an unfalsifiable condition into a checkable one. A party who declares hazard and discloses the existence of the work, its questions, its decisions and its interpretations has behaved as a party with a genuine hazard would. A party who declares hazard and discloses nothing has behaved as a party would who wanted silence and reached for the strongest available reason.

12.4 Patterns over Time

Individual restrictions are frequently ambiguous and patterns are not.

A party whose restrictions are always on the same ground, always without a schedule, always at the narrowest class, and always without a residue, has a practice that is visible in the record even where no single restriction can be faulted.

Claim 12.3. *The record makes a party's practice of restriction inspectable, and inspection of the practice is more informative than inspection of any restriction within it. This is the principal contribution the design makes to the governance of disclosure, and it is a contribution to visibility and not to control. This claim is formal.*

12.5 The Limits

Three, and the third is the general condition of the series.

Detection is not remedy. A reader who determines that a ground was misapplied has learned something and acquired no means of altering it, and the account supplies no procedure by which a restriction is overturned.

Detection depends on someone looking. The record makes a practice inspectable and does not make it inspected, and the parties best placed to look are frequently those with the least reason to.

And a community that does not care will not be made to care by a record. Every mechanism in this account produces information for parties who wish to use it, and none produces the wish.

Claim 12.4. *The account contributes visibility of restriction and contributes no enforcement, no remedy, and no reason for a community to attend. Recorded under the standard of the companion paper on incentives, the governance of restriction is addressed at the level of information and unaddressed at the level of consequence. This claim is formal.*

12.6 Summary of Results

This section has treated detection. The four named failures are assembled, and Claim 12.1 establishes that detectability is inversely related to the width of the restriction a ground supports, so hazard, which supports the widest, is least detectable and must be policed by a body a community constitutes rather than by readers. Claim 12.2 establishes the residue as the observable available for every ground, turning an unfalsifiable condition into a checkable one, since a declaration with its residue costs the declaring party something and a declaration without it costs nothing. Claim 12.3 establishes that a party's

practice of restriction is more informative than any restriction within it, which is the design's principal contribution to this governance and is a contribution to visibility alone. Claim 12.4 records the limits: detection is not remedy, depends on someone looking, and produces no reason for a community to attend.

13 Jurisdictional Variation and the Separation of Mechanism from Policy

This section treats an arrangement whose participants are subject to different legal regimes. Its role is to state how such an arrangement operates without the account taking a position on any question of law. Its structure treats the problem, the separation of levels, what law constrains structurally, the fields an implementation must carry, and the limits. Its method is analysis of a design constraint, and the section adjudicates no legal question and states no legal conclusion.

13.1 The Problem

Participants in one trajectory may be in several jurisdictions. Recording consent, retention obligations, rights of erasure, restrictions on transfer, and the treatment of research material differ among them, and the differences are not reconcilable by choosing the strictest, since obligations in different regimes may conflict rather than nest.

An arrangement operating across such regimes must be capable of behaving differently in each without the specification of the arrangement changing.

13.2 The Separation of Levels

The companion papers fix four levels: external law, the operating norms a community adopts, the protocol, and applications. The separation is what makes cross-jurisdictional operation expressible.

Claim 13.1. *Law belongs to a level the account describes and never encodes. A specification encoding the rules of any jurisdiction would be wrong somewhere immediately and obsolete everywhere within a few years, since the rules change independently of any design and differ among regimes that must interoperate. This claim is formal.*

Claim 13.2. *Law nonetheless reaches the design, and it reaches it negatively. A data model incapable of expressing redaction forecloses lawful operation in every regime with a right of erasure, and no charter repairs it. Structural incapacity is therefore the channel through which law constrains a specification, and it is the only such channel; that a technical configuration establishes rules of an order comparable to law is the general form of the point (Reidenberg, 1998; Lessig, 2006). This claim is formal.*

13.3 The Structural Constraints Imposed by Law

Four capacities follow from the second claim and are stated as design properties in place of legal conclusions.

Separability. Personal content must be separable from the attested skeleton, so that removal of content leaves the record's structure and its verification intact, which is the reconciliation the literature on erasure rights and append-only records converges on (Finck, 2018; European Union, 2016).

Expressibility of redaction. An arrangement must be able to record that content was removed, when, by whom and on what ground, without removing the record of the removal.

Consent as a recorded object. A record concerning a person must be able to reference a consent whose scope and date are recorded, since regimes differ in what consent must cover and an arrangement unable to reference one cannot comply with any of them.

Locality of storage. An arrangement must be able to record where material is held and to restrict where it may be replicated, since regimes differ in what may cross which borders.

Claim 13.3. *These four are capacities and not rules. An arrangement possessing them may operate lawfully in a regime and may also operate unlawfully; an arrangement lacking them cannot operate lawfully in regimes that require what they express. The design's obligation is to possess the capacities, and the determination of lawfulness lies with implementers and their advisers. This claim is formal.*

13.4 The Location of the Rules

The rules themselves live in the operating charter, and the requirement on it is stated here.

Requirement 13.4. (Charter statement of legal operation). *A charter must state the jurisdictions under which the arrangement operates, the consent required before a record concerning a person is created, the retention applying to each class of record, the treatment of erasure requests, and any restriction on where material may be held. A charter that leaves these unstated leaves them to whoever operates the implementation.*

An arrangement spanning regimes will accordingly hold several charters, or one charter with regime-specific provisions, and a record must identify which charter and version governed it when it was created.

13.5 The Limits

Four.

Federation distributes the compliance burden and does not remove it. Each implementation is subject to the regimes reaching it, and an implementation receiving records from another inherits obligations it did not create.

Conflicting obligations are not resolved by this account. Where a regime requires retention and another requires erasure of the same material, the conflict is real, and an implementation resolves it in a way it records, with no support from the specification.

Records already propagated are not reached. A redaction notice reaches implemen-

tations that receive it and reaches no copy held by a party who has departed.

And an arrangement cannot make its participants' obligations disappear. A participant subject to an obligation about material they hold remains subject to it, whatever the charter says about the arrangement.

Claim 13.5. *The account supplies capacities, requires a charter to state rules, and resolves no conflict of laws. Where obligations conflict, the arrangement records what was done and the resolution rests with parties the account does not govern. This claim is formal.*

13.6 Summary of Results

This section has treated jurisdictional variation. Claim 13.1 establishes that law is described and never encoded, since an encoding would be wrong somewhere at once and obsolete everywhere soon. Claim 13.2 establishes that law reaches the design negatively, through structural incapacity, which is the only channel. Four capacities follow as design properties: separability, expressibility of redaction, consent as a recorded object, and locality of storage, and Claim 13.3 distinguishes possessing a capacity from operating lawfully. Requirement 13.4 obliges a charter to state jurisdictions, consent, retention, erasure treatment and locality, on the ground that unstated rules fall to the operator. Four limits are recorded: federation distributes the compliance burden, conflicting obligations are unresolved, propagated copies are unreachable, and participants' own obligations are untouched.

14 Predictions and the Conditions of Falsification

This section states what the account predicts and what would show it wrong. Its role is to make a normative typology testable, which is unusual and is possible here because the account makes claims about what parties will do. Its structure treats the predictions, the refuting observations, the observation that would show the set of grounds incomplete, and the measurement difficulties. Its method is enumeration, with each prediction derived from a named claim.

14.1 The Predictions

P1. Restrictions declared on the hazard ground will withhold more than the ground covers, more often than restrictions on the other three, since hazard supports the widest restriction and its misapplication is least detectable. Derived from Claim 12.1.

P2. Where residues are required and inspected, declarations of appropriability will fall, since a declaration accompanied by its residue costs the declaring party something. Derived from Claim 6.4.

P3. Arrangements will accumulate restricted material over time, since only the vulnerability ground admits a recorded terminus and the other three do not. Derived from Claim 7.4.

P4. Communities will declare vulnerability where appropriability obtains, since vulnerability is the easiest ground to assert and carries no admission of commercial interest.

Derived from the limits recorded in Section 5.

P5. Displacement of exploration into simulation will be followed by contestation over benchmarks, and the contestation will concern technical specification in place of allocation. Derived from Claim 8.2.

P6. Third-party assembly of trajectory corpora will occur wherever a substantial corpus exists, and will occur without compensation to the producing parties. Derived from Claim 10.1 and 10.3.

P7. Where an arrangement records restrictions and nobody inspects them, restriction practice will not differ from practice in arrangements that record nothing. Derived from Claim 12.4.

14.2 The Refuting Observations

Hazard restrictions that disclose their residues as fully as appropriability restrictions do refute P1 and the detectability ordering.

Appropriability declarations unchanged in frequency where residues are required and inspected refute P2 and, with it, the argument that the residue requirement does any work.

Arrangements whose restricted proportion falls over time without a change of ground refute P3.

Communities declaring appropriability where it obtains, in preference to the easier ground, refute P4.

Displacement into simulation without contestation over benchmarks, sustained over a period in which allocation would have been contested, refutes P5 and the displacement claim.

Substantial corpora existing without third-party assembly refute P6.

And restriction practice differing measurably between recording and non-recording arrangements, where nobody inspects the records, refutes P7 and would show that recording alters behaviour independently of inspection.

14.3 The Observation That Would Show the Set Incomplete

The account closes the set of grounds, and the closure is the strongest thing it asserts.

Claim 14.1. *A restriction that a competent and disinterested reader judges reasonable, and that fits none of rivalry, hazard, exploratory vulnerability and appropriability, would show the set incomplete. The account invites the observation, since a typology whose exhaustiveness cannot be challenged is not a typology but a stipulation. This claim is formal.*

Two candidates are worth naming, since they are the ones a reader will reach for.

Restriction to protect a third party who is a subject of the work, in medical or social research, appears at first to fit none of the four. On the account offered here it is hazard, with the propagation of the content creating a risk to an identified party in place of a general one, and a reader who finds that reading forced has found a candidate.

Restriction to honour an undertaking given to another party, such as a confidentiality condition attached to data received, appears to fit appropriability and does not always: the excludability may protect the party who supplied the data and not the party who received it. This is the stronger candidate, and the account's provisional answer is that the receiving party declares appropriability on behalf of the supplying party's condition, which is strained.

14.4 The Measurement Difficulties

Three.

Restrictions in arrangements without a record are unobservable, so the predictions are testable only in arrangements conforming to something like the companion specification.

Whether a residue was disclosed is observable, and whether a residue was complete is not, since a reader cannot know what was withheld.

And the misapplication predictions concern the relation between a declaration and a condition that only the declaring party can assess, so they are testable in aggregate patterns and not in individual cases.

14.5 Summary of Results

This section has stated seven predictions with their derivations and refuting observations. Claim 14.1 states the observation that would show the set of grounds incomplete, and two candidates are examined: protection of a research subject, treated here as hazard with an identified party, and confidentiality undertakings to a supplying party, whose treatment as appropriability is strained and which is the stronger candidate. Three measurement difficulties are recorded: restrictions in non-recording arrangements are unobservable, the completeness of a residue is unobservable, and misapplication is testable in aggregate patterns and not in individual cases.

15 The Boundaries of the Account and Its Conclusion

This section states what the paper has shown, what it leaves unsettled, and what it declines to claim. Its role is to close with the limits in one place. Its structure treats the results, the unsettled matters, the refusals, and the conclusion. Its method is enumeration.

15.1 The Results

Six.

Restriction in an arrangement of this kind is justified by a reason and not by a title, since the instruments that confer titles protect finished products and a scheme of exclusive rights over trajectory content would install the convergence the series diagnoses.

A ground determines an object and a residue. Four grounds are distinguished, they are irreducible to one another, and each restricts a different object: rivalry restricts a re-

source, hazard the propagable content of a method, vulnerability a time, appropriability the content whose disclosure destroys excludability.

The residue requirement is what prevents a ground from operating as a licence, and it supplies the only observable available to a reader in every case, including the case of hazard where the condition itself cannot be assessed from outside.

Displacement of exploration into simulation relocates the governance problem from the allocation of a resource to the specification of an environment, and the relocation is unfavourable in one respect: allocation is visible and contestable where specification is continuous and technical.

The component of a trajectory that carries its value in reuse is the component that carries its evidential force and its hazard, so no design publishes the trajectory minus its dangerous parts and retains the benefit, and the account is accordingly a typology of grounds that separates parties in place of a rule that separates content.

And downstream use of records by parties who did not produce them is governed by none of the four grounds, since all four concern disclosure and this concerns subsequent use.

15.2 The Matters Left Unsettled

Five.

The closure of the set. Confidentiality undertakings given to a party who supplied material fit appropriability only with strain, and this is the observation most likely to show the set incomplete.

The assessment of hazard. The account requires the ground to be declared and supplies no competence to assess it, and the assessment cannot be performed by parties outside the restricted class.

The terminus of restrictions on three of the four grounds. Only vulnerability admits a recorded terminus, so arrangements accumulate restricted material, and nothing in the account obliges release when a condition ceases.

The bound on the restricting authority. Portability, which bounds every other position the series treats, bounds this one weakly, since a departing party cannot take what they were never shown.

And the terms of downstream use. A charter is required to state them, they are difficult to enforce, and the parties who bear the cost of producing records are least placed to bring the action that would enforce them.

15.3 The Claims Declined

Three.

It does not claim that the four grounds are the correct set, only that they are irreducible to one another and that the account invites the observation that would show them incomplete.

It does not claim that the apparatus prevents improper restriction. It supplies a vocabulary in which a restriction is an assertion, a named failure for each ground, and one

observable, and it supplies no enforcement, no remedy and no reason for a community to attend.

And it does not claim that openness is preferable to restriction. Each ground states a condition under which restriction is legitimate, and the appropriability ground is treated as a condition of some work existing at all rather than as a concession.

15.4 Conclusion

The question the paper asks is narrow and its answer is a discipline rather than a mechanism.

An arrangement that records how understanding changes must withhold, and the question is what follows from why. The account's contribution is that a reason, once declared, determines an object and leaves a residue, so that a restriction becomes a dated attributable assertion whose adequacy a reader can check by asking a single question: was what the ground leaves disclosable in fact disclosed.

That is a modest contribution and it is the one available. The mechanisms that would do more all require an authority to assess conditions that only the restricting party can assess, and such an authority is a position of exactly the kind the series spends its length constraining. The account therefore stops at visibility, states that visibility without inspection changes nothing, and leaves the inspecting to communities that may or may not perform it.

What the paper is most confident of is negative. No design separates the valuable from the dangerous within a trajectory, since they are the same material; no ground supports a restriction wider than its object; and no arrangement that withholds its residues is withholding for the reason it declares.

15.5 Summary of Results

This section has stated six results: restriction justified by reason in place of title; four irreducible grounds each determining an object and a residue; the residue requirement as the barrier against a ground operating as a licence and as the only universally available observable; the displacement of the governance problem by simulation; the shared component of value, evidence and hazard; and the exclusion of downstream use from all four grounds. Five matters are left unsettled: the closure of the set, the assessment of hazard, the absence of a terminus on three grounds, the weak bound on the restricting authority, and the enforcement of downstream terms. Three claims are declined: that the set is correct, that the apparatus prevents improper restriction, and that openness is preferable to restriction. The conclusion states that the contribution is a discipline in place of a mechanism, and that the paper's most confident results are negative.

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